How-To Library

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Creating a JMX Logger in Dynamic Data Masking

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Abstract

This article describes how to use JMX to view Dynamic Data Masking Server event logs in JConsole.

Supported Versions

- Dynamic Data Masking 9.7.0 or later

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Overview

Create a custom appender in the Management Console to log Dynamic Data Masking Server events in JConsole.

In the Dynamic Data Masking Management Console, logger nodes define the events that you want to log and appender nodes define how to log the events.

The Management Console tree contains system loggers, which are pre-defined logger nodes that log Dynamic Data Masking Server, service, and rule events. You can create custom loggers that you can use in security rules to define specific events that you want to log.

Appenders use log4j classes to define the output format of the log information. The system loggers contain appenders that create the audit trail, rule, and server logs. You can add appenders to create rolling file, SNMP, SMTP, Syslog, and custom log outputs.

In this example, you create a custom appender under the rootLogger system logger node. Because the appender is under the rootLogger node, the appender logs Dynamic Data Masking Server events. The rootLogger logger also sends the information to the server.log file. The appender uses the Custom appender type to create a JMX log output that you can view in JConsole.

In this example, you complete the following tasks:

1. Download the JMX jar files.
2. Restart the Dynamic Data Masking Server.
3. Create the JMXLogger appender.
4. Subscribe to the JMX appender in JConsole and view the log information.

Step 1. Download the JMX Jar Files

Download the JMX jar files and save the files in the Dynamic Data Masking installation directory.

Save the jar files in the following location:

```
<Dynamic Data Masking installation>/jre/lib/ext
```

Save the following files in the Dynamic Data Masking installation directory:

jmx-logger.jar

Download the jmx-logger.jar file from http://www.java2s.com/Code/Jar/j/Downloadjmxloggerjar.htm.

mvel2-2.2.0.Final.jar

Download the mvel2-2.2.0.Final.jar file from http://mvnrepository.com/artifact/org.mvel/mvel2/2.2.0.Final.

log4j-1.2.17.jar

Download the log4j-1.2.17.jar file from http://mvnrepository.com/artifact/log4j/log4j/1.2.17.

Step 2. Restart the Dynamic Data Masking Server

Restart the Dynamic Data Masking Server before you create the JMX appender.

Open a Server Control window and run the following commands:

- `server stop`
- `server start -Dcom.sun.management.jmxremote`

After you restart the Dynamic Data Masking Server, log in to the Management Console.

Step 3. Create the JMXLogger Appender

Create a custom JMXLogger appender in the Management Console.

Note: You can create a custom logger and add the JMX appender to the logger. In this example, the appender is under the rootLogger node and logs server events and error messages.

1. In the Management Console tree, expand the Loggers node and the rootLogger node.
2. Select the rootLogger node and click **Tree** > **Add Appender**.

   The **Add Appender** window appears.
3. Select the **Custom** appender type.
4. Configure the following appender parameter:

   Name

   Enter the name of the appender that appears in the Management Console tree. In this example, the appender name is JMXLogger.
5. Click the plus sign (+) to add lines in the **Properties** box and enter the following property names and values:

   MBeanServer

   Enter the IP address or host name of the MBeanServer that you use to view JMX messages. In this example, the MBeanServer IP address is 127.0.0.1.

   ObjectName

   Enter the following parameter value:

   ```
   <object name>:type=LogEmitter
   ```

   The object name is the JMX MBean name for the JmxLogger object.

   layout

   Enter the following parameter value:

   ```
   org.apache.log4j.PatternLayout
   ```

layout.ConversionPattern

Enter the following parameter value:

```
%d{MM/dd HH:mm:ss,SSS} [%t] %-5p - %m%n
```

The layout.ConversionPattern value is the pattern in which JMX prints the logs. For more information about the PatternLayout class, go to https://logging.apache.org/log4j/1.2/apidocs/org/apache/log4j/PatternLayout.html.

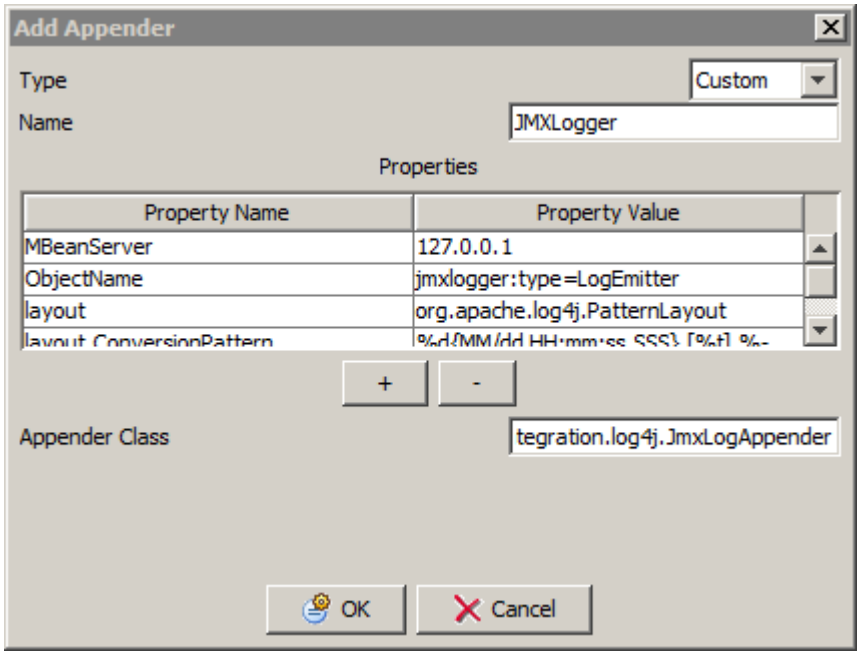
6. Configure the following appender parameter:

AppenderClass

Enter the following text:

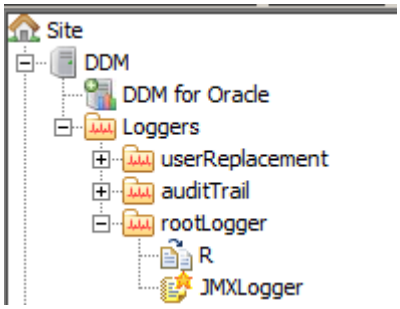
```
jmxlogger.integration.log4j.JmxLogAppender
```

7. Review the appender parameters and values. The following image shows the Add Appender window with the parameter values:



8. Click **OK** to save the appender.

The appender appears in the Management Console tree. The following image shows the Management Console Tree:



Step 4. Subscribe to the JMX Appender

Subscribe to the JMX appender to receive Dynamic Data Masking Server event notifications.

This example uses JConsole to view the appender notifications. For more information about how to use JConsole, go to http://docs.oracle.com/javase/7/docs/technotes/guides/management/jconsole.html.

1. In a command prompt, navigate to `JDK_HOME/bin` and start JConsole with the `jconsole` command.

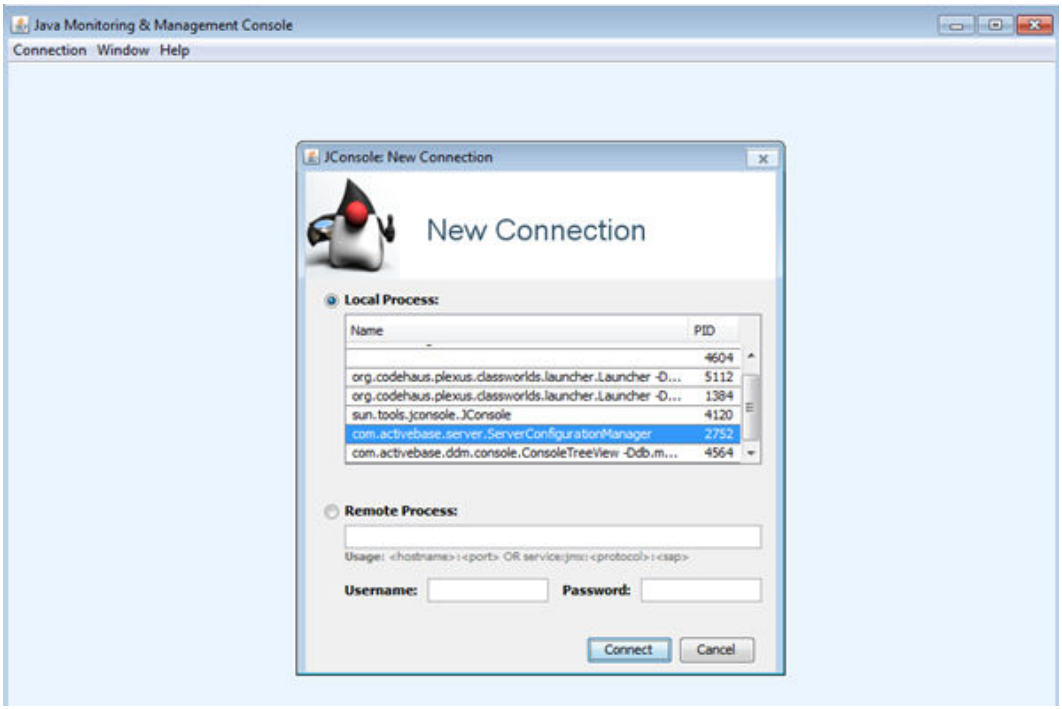
   For example, you might enter the command in the following location:

   ```
   C:\Program Files\Java\jdk1.8.0_60_bin>jconsole
   ```

2. In the **New Connection** window, select the following connection:

   ```
   com.activebase.server.ServerConfigurationManager
   ```

   The following image shows the **New Connection** window:

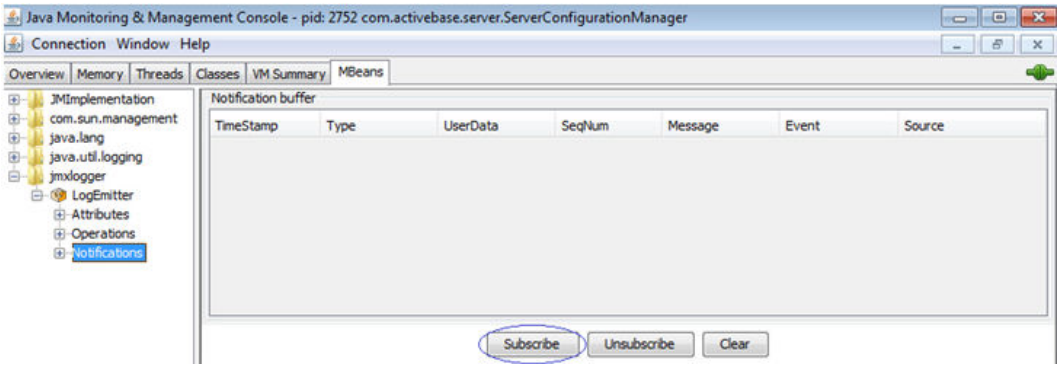


3. Click **Connect**.
4. Click the MBean tab to view the JMX appenders. Expand the node with the name of the appender that you created in the Management Console and expand the LogEmitter node below the appender node.

   Below the LogEmitter node, there are Attributes, Operations, and Notifications nodes.

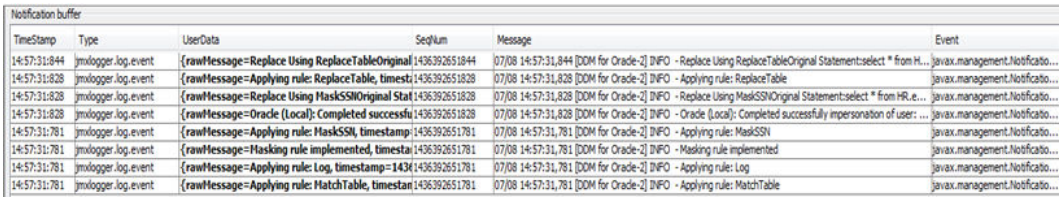
5. Select the Notifications node and click **Subscribe**.

   The following image shows the appender node and the **Subscribe** button:



6. After you subscribe to the appender notifications, the notifications appear in the JConsole Notifications window.

   The following image is an example of the appender notifications:



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Acknowledgements

Thanks to Prashant Jain and the Dynamic Data Masking development team for their help in completing this article.